

GLOBULAR CLUSTER

M12



CATALOG NUMBERS
M12, NGC 6128

DISTANCE FROM SUN
16,000–18,000 light-years

MAGNITUDE 7.7

OPHIUCHUS

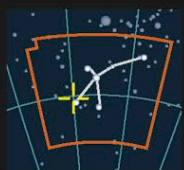
Discovered by Charles Messier (see p.73) in 1764, M12 was one of the first globular clusters to be recognized. M12 is at the very limit of naked-eye visibility and therefore best viewed with a telescope. The cluster contains many bright stars and is condensed toward the center. Its stars are spread across a distance of about 70 light-years, making it less compact than most. Because of this, M12 was originally regarded as an intermediate form of cluster, between open clusters and globular clusters, before the two types were recognized as being fundamentally different.



EARLY DISCOVERY

GLOBULAR CLUSTER

NGC 4833



CATALOG NUMBER
NGC 4833

DISTANCE FROM SUN
17,000 light-years

MAGNITUDE 7.8

MUSCA

NGC 4833 is a small globular cluster in the southern constellation Musca and therefore is not visible to most observers in the Northern Hemisphere. It was discovered by Nicolas Louis de Lacaille (see p.422) during his 1751–1752 journey to South Africa. Although NGC 4833 is too faint to see with the naked eye, it is easily visible through a small telescope. However, because the cluster is rich and compact, even a moderate amateur telescope fails to resolve its stars fully. The center of the cluster is only slightly more dense than its surroundings, and consequently the cluster lacks the gravitational pull needed to hold onto its stars and many have already left the cluster. NGC 4833 is located below the galactic plane behind a dusty region. The dust absorbs light from the cluster and causes

its starlight to redden. Because of this reddening, astronomers studying this globular cluster have had to correct the apparent magnitudes of the various stars being studied. The technique used is applied to all globulars lying near the galactic plane. The cluster contains at least 13 confirmed RR Lyrae variable stars (see pp.282–283), which have helped astronomers to estimate the cluster's age at about 13 billion years.

COMPACT CLUSTER

NGC 4833 was first recorded by Nicolas Louis de Lacaille in 1752 as resembling a comet. However, with modern, high-powered telescopes, it is seen as a well-resolved and compact cluster with a scattering of outlying stars.

DISTANT GLOBULAR

GLOBULAR CLUSTER

M14



CATALOG NUMBERS
M14, NGC 6402

DISTANCE FROM SUN
23,000–30,000 light-years

MAGNITUDE 8.3

OPHIUCHUS

The globular cluster known as M14 has a diameter of about 100 light-years and contains several hundred thousand stars. Because of its considerable distance, it is too faint to be seen with the naked eye and, although binoculars or a small telescope will reveal the cluster, a larger instrument is needed to resolve individual stars. Many amateur observers mistakenly identify this object as an elliptical galaxy. In 1938, M14 was home to the first nova photographed in a globular cluster. However, subsequent searches with some of the world's most powerful telescopes have failed to find either the nova star or any of its remnants.

GLOBULAR CLUSTER

M107



CATALOG NUMBERS
M107, NGC 6171

DISTANCE FROM SUN
27,000 light-years

MAGNITUDE 8.9

OPHIUCHUS

A relatively “open” globular cluster lying close to the galactic plane, M107 is too faint to be seen with the unaided eye. Observations through large telescopes have revealed that the cluster contains dark regions of interstellar dust that obscure some of its stars. This is quite unusual in globular clusters. M107 spans a distance of about 50 light-years.



LOOSE CLUSTER

GLOBULAR CLUSTER

M68



CATALOG NUMBERS
M68, NGC 4590

DISTANCE FROM SUN
33,000–44,000 light-years

MAGNITUDE 9.7

HYDRA

M68 is a globular cluster that is visible only through telescopes. It appears as a small patch when viewed with binoculars, but small telescopes can reveal its constituent stars and its densely populated center. The cluster has a diameter of about 105 light-years, and its orbit around the center of the Milky Way means that it is approaching the solar system at about 250,000 mph (400,000 kph). Although many variable stars (see pp.282–283) have been detected within the cluster—more than 40 to date, including RR Lyrae stars—the distance to M68 is still uncertain.



DENSE BALL

GLOBULAR CLUSTER

M15



CATALOG NUMBERS
M15, NGC 7078

DISTANCE FROM SUN
35,000–45,000 light-years

MAGNITUDE 6.4

PEGASUS

At the limit of naked-eye visibility, M15 is one of the densest globular clusters in the Milky Way. The cluster has a diameter of about 175 light-years, but, as the center of the cluster has collapsed in on itself, half of its mass is located within its 1-light-year-wide superdense core. M15 also contains nine pulsars, remnants of ancient supernova explosions (see pp.266–267) from

PACKED CORE

At its core, this globular cluster has the highest concentration of stars in the Milky Way outside the galactic center.

the time when the cluster was young. Unusually, two of these pulsating neutron stars form a contact binary pair (see p.274).

TRUE COLORS

The brightest stars in M15 are red giants, with surface temperatures lower than the Sun's. Most of the fainter stars are hotter, giving them a bluish-white tint.

